



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Mechanical Engineering
Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: V Semester B.E Mechanical Engineering

Course Code & Title: CME380 Automobile Engineering

Name of the Faculty member (s): Mr.L.Karthikeyan

Innovative Practice Description

- **Unit / Topic:** Unit II/ Vehicle Structure and Engines
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO 1
- **Activity Chosen:** Reflective Journal activity
- **Justification:**
 - A reflective journal activity is a process where students present their learnings and inferences from recent research articles which relevant to the course content.
 - The primary objective to encourage student's self- reflection and exploration of one's work and scientific report (Relevant to the course knowledge)
- **Time Allotted for the Activity:** 10 minutes / Team
- **Details of the Implementation:**
 1. The students are instructed to form a team of three members and allowed to visit library on 28.06.2023 during 7th hour (03.00PM to 04.00PM)
 2. RIT central library having subscription of "International Journal of Vehicle structure and systems" which is most relevant to the course content
 3. The teams are instructed to infer anyone recent research article published in said journal as per their own interest
 4. Finally, the team presented their inferences in selected article and evaluated as per below rubrics

Criteria	Poor (1-2 Marks)	Good (3-4 Marks)	Excellent (5 Marks)
Problem Statement / Introduction (5 Marks)	<i>Did not infer the article properly</i>	<i>Presented the research topic and the need</i>	<i>Clearly defined the context of research</i>
Coverage of content (5 Marks)	<i>Major works omitted / significance to article not presented</i>	<i>Major works included but not covered in adequate depth</i>	<i>Appropriate resources covered in depth / significance of work critiqued</i>
Communication (5 Marks)	<i>Delivery is poor / unclear language and appears uncomfortable</i>	<i>Delivery makes the presentation interesting and appears comfortable</i>	<i>Delivery makes the presentation compelling and confident</i>
Clarity of presentation (5 Marks)	<i>Article not presented clearly / Not in sequence</i>	<i>Presentation is clear but not concise</i>	<i>Presentation is clear and concise / coherent</i>



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

- CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO10	PSO1
CO1	3	2	2	2	2

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO10	PSO1
	3	2	2	2	2
Justification for correlation	High correlation	Review literature	Appropriate consideration for Environment	Students presentation Evaluation as per rubrics	Understanding Vehicle construction

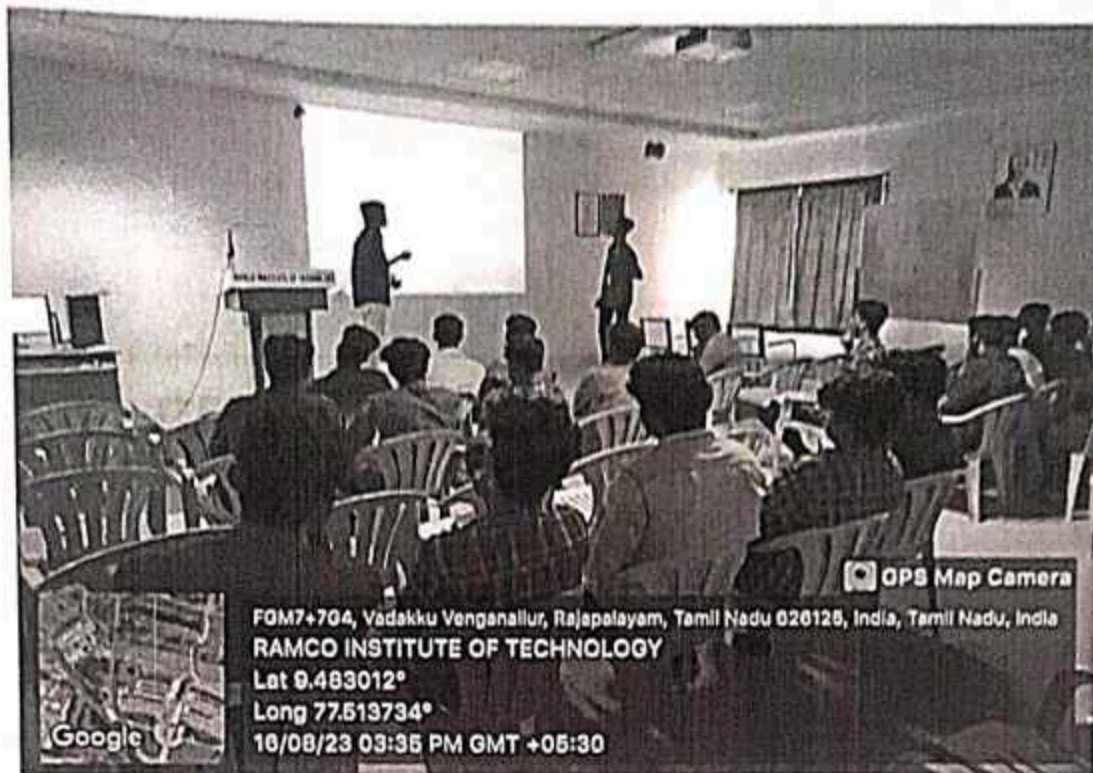
- Images / Screenshot of the practice:





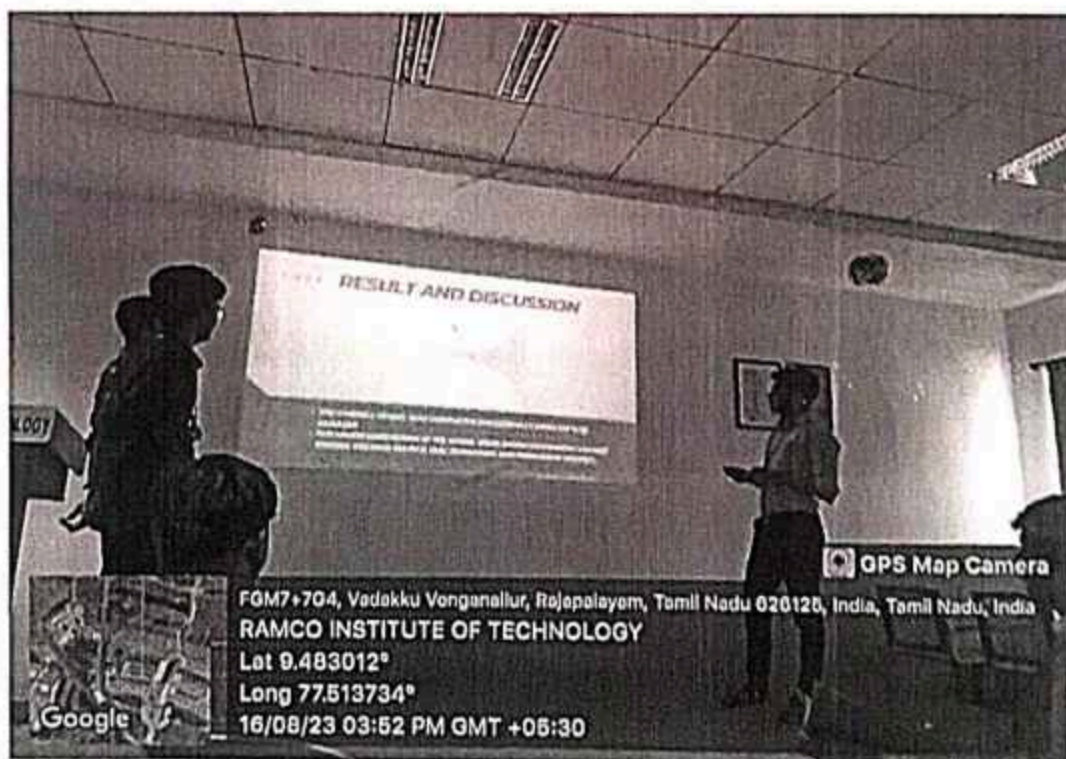
RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH



Google

FOM7+704, Vadakku Venganallur, Rajapalayam, Tamil Nadu 626125, India, Tamil Nadu, India
RAMCO INSTITUTE OF TECHNOLOGY
Lat 9.483012°
Long 77.513734°
16/08/23 03:35 PM GMT +05:30



Google

FOM7+704, Vadakku Venganallur, Rajapalayam, Tamil Nadu 626125, India, Tamil Nadu, India
RAMCO INSTITUTE OF TECHNOLOGY
Lat 9.483012°
Long 77.513734°
16/08/23 03:52 PM GMT +05:30



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Based on the written feedback collected from the students, they expressed their satisfaction about this activity, motivation towards self-learning and improving technical presentation skills.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ It's a task that encourages students to apply fundamental ideas to higher level thinking.

- ✓ It also helps the students to develop research and critical thinking skills while gaining an understanding of the current state of knowledge

- ✓ The students can understand to synthesize the information they gather from the literature / Research article and compile their conclusions into a well-structured presentation

- ✓ Most of the students reported that, this activity helped them to overcome stage fright

- ❖ **Challenges faced in implementation:**

- Absenteeism of students due to personal reasons

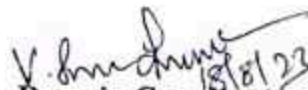
- Venue availability & Additional hours used for completing the presentations

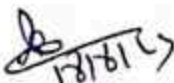
References:

- ❖ <https://www.niu.edu/citl/resources/guides/instructional-guide/reflective-journals-and-learning-logs.shtml>

- ❖ <https://maftree.org/cja/index.php/ijvss/index>


Signature of Faculty Member


Domain Coordinator


HOD

Activity Sheet I - Foundational to higher-order thinking

Date: 28.07.2023

Sl.No	Student Name	Journal Title	Presentation topic	Signature
1.	Hanish Krishnan R	International Journal of Vehicle structure & Systems	A Review of wheel - Rail Dynamic Behaviour on Curved Tracks	R. Hanish
	Jeya Prakash K			R. Jeya
	Muthu Kumar P			R. Muthu
2.	Harishankumar T	International Journal of Vehicle structure & Systems	Study of Stochastic optimization of Post weld heat treatment of dissimilar TIG weldment of P355 steel	R. Harishankumar
	Vignesh B			R. Vignesh
3.	Gowtham J	International Journal of Vehicle structure & Systems	A survey on internet of vehicles application, localization, communication model and challenges	J. Gowtham
	Ganesh kumar P.			P. Ganesh
	Subbiah Selvam M.			M. Subbiah
4.	M-Preem Sukumar	International Journal of Vehicle structure & Systems	Development of an advanced vehicle restraint system with design and development of electromagnetic suspension for development of an advanced vehicle restraint system with an adaptive steering column for special consideration of small occupants	M. Preem
	I. Saravana Kumar			I. Saravana
	V. Suvyaskumar			V. Suvyaskumar
5.	R- Madhavan	International Journal of Vehicle structure & Systems		R. Madhavan
	R- Ramshanker			R. Ramshanker
	M - Suvyaskumar			M. Suvyaskumar

Sl.No	Student Name	Journal Title	Presentation topic	Signature
6.	Karthick K	International Journal of Vehicle structure & Systems	STUDY ON VIBRATION CONTROL ON INDIAN TRACTORS	Shruti
	Shridharan R			SP
	Manikanda Prabhu			Mani
7.	A. Balaji	International Journal of Vehicle structure & Systems	IMPACT OF STEERING STIFFNESS ON The Injury Criteria	MR.
	P. Anandhan			RA
	Sateesh Bharathi			AB
8.	A. Dinesh	International Journal of Vehicle structure & Systems	INFLUENCE OF HAND MATERIAL COATING ON THE AUTOMOTIVE BRAKE LINEAR	He
	S. Saranya Kumar			South
	P. Sudarshan			Selvar
9.	M. Aathithyan	International Journal of Vehicle structure & Systems	Numerical Analysis of Birel Impact over the Aircraft wing shield	M. Aathi
	S. Mohamed Anish			S. Anish
	-			-
10.	P. Sivaprasakash	International Journal of Vehicle structure & Systems	COMBUSTION CHARACTERISTICS OF MULTICOMPONENT FUEL DROPLET	Prasath
	S. Surya			S. Surya
	C. Palpani			C. Palan
11.	I. Ramkumar	International Journal of Vehicle structure & Systems	Crash Dependency of severe plastic deformation during combined Extrusion, Forging	U. Ram
	N. Balakumar			Arul
	M. Duryaany			Arka

COI: Recognize the various parts of the automobile and their functions and materials

A. H. 10/12/23
Faculty I/C

S. S. 28/11/23
Library I/C

V. Anandhan 11/8/23
Domain coordinator

11/8/23
HOD/MECH

A Survey on Internet of Vehicles Applications, Architecture, Communication Models and Challenges

T. Sridevi^a, S. Cloudin^{a,c} and P. Mohankumar^b

^aKCG College of Tech., Chennai, Tamilnadu, India

^bSRR Engg. College, Chennai, Tamilnadu, India

^cCorresponding Author, Email: scloudin@kcgcollege.com

ABSTRACT:

IoT refers to connecting multiple objects (cars, home appliances etc.) to internet and they are capable of sending and receiving electronic information through sensors and APIs. Internet of Vehicles (IoV) is an application of IoT, which involves connecting vehicles to internet to provide intelligent transportation, smart and safe driving in private/public vehicle. In this paper we are projecting the applications of IoV, different architecture models, communication interaction models and challenges in IoV. This paper would be helpful for researchers to further explore the potentials of IoV.

KEYWORDS:

Internet of Vehicles; Communication models; System architecture

CITATION:

T. Sridevi, S. Cloudin and P. Mohankumar, 2021. A Survey on Internet of Vehicles Applications, Architecture, Communication Models and Challenges, *Int. J. Vehicle Structures & Systems*, 13(3), 304-308. doi:10.4273/ijvss.13.3.12.

1. Introduction

In the revolutionizing era of Internet of Things (IoT), concepts of intelligent transportation system, smart driving, safe driving in transportation industry is achieved through seamless integration of connecting vehicles to Internet and thus forming Internet of Vehicles (IoV). IoV has started gaining recognition by researchers, practitioners, industries, standards and governments, at present it is in the stage of evolving concepts into reality. Multiple researches in the segment of IoV is still going on and the progress is inevitable. Experts forecast that by the year 2020 there will be total of 50 billion devices connected to the internet [1]. Gartner predicts that by 2020, more than 250 million vehicles will be connected globally [2]. The IoT market is separated by applications into purchaser gadgets, production, transportation, medical services, retail and others. Among these sections, the transportation gets the highest investment closely followed by the consumer electronics segment [37].

In this paper applications of IoV are categorized based on safety, efficiency, comfort and infotainment. Architecture models proposed by various authors have been discussed. Communication interaction models has been categorized as i) intra-vehicle and ii) inter-vehicle models. Vehicle-to-Sensors (V2S), Vehicle-to-Driver (V2D), Device-to-Device (D2D) falls under intra-vehicle interaction model while Vehicle-to-Vehicle (V2V) and Vehicle-to-Roadside (V2R) falls under inter-vehicle interaction model. Challenges in implementing IoV has also been discussed.

2. Applications

Multiple authors have proposed applications in the view of considering the safety, traffic efficiency and infotainment. Considerable amount of few applications is surveyed. The applications of IoV have been classified as a.) safety-based, b.) efficiency based, c.) comfort based and d.) infotainment based.

2.1. Safety based

They are generally known as Collision Avoidance System (CAS) and their primary responsibility is to identify any potential situation of collision within the transportation system and avoid them. There are many studies which outlines different approaches for cooperative CAS, to name a few. An approach using V2V for cooperative collision avoidance at intersections was proposed [3]. Decentralized algorithms were used to achieve CAS. Same was tested with automated vehicles and it worked effectively. C-RACCA (cluster-based risk-aware cooperative collision avoidance) and this potentially identifies a collision scenario within a cluster of vehicles in the same network [4]. Cluster of networks was achieved through following ways fixing GPS, establishing wireless channels to forward warning messages, relative velocity with respect to neighbouring vehicles is estimated, on arrival of warning message, vehicle could estimate the origination direction either from front or rear and slowly decelerates considering other vehicles at front/back also decelerates, also vehicles have knowledge about its wireless transmission range. This simulation reduced the number of collisions. Proposal on Smart eye [5] plays key role in accident

Team No: 8

Student Name 1: P. Ganesh kumar

Student Name 2: J. Goutham

Both performed well.

Student Name 3: M. Subbiah @ selvam (01)

Criteria	Poor (1-2 Marks)	Good (3-4 Marks)	Excellent (5 Marks)	Marks awarded
Problem Statement / Introduction (5 Marks)	Did not infer the article properly	Presented the research topic and the need	Clearly defined the context of research	5
Coverage of content (5 Marks)	Major works omitted / significance to article not presented	Major works included but not covered in adequate depth	Appropriate resources covered in depth / significance of work critiqued	5
Communication (5 Marks)	Delivery is poor / unclear language and appears uncomfortable	Delivery makes the presentation interesting and appears comfortable	Delivery makes the presentation compelling and confident	5
Clarity of presentation (5 Marks)	Article not presented clearly / Not in sequence	Presentation is clear but not concise	Presentation is clear and concise / coherent	5
Total				20

Student feedback for the reflective journal activity:

we gained knowledge about new technology
we gained information and skills to use the journals
we also gained knowledge about the Internet of vehicles
and Internet of Things

D. H. 12/12/23
Panel Member 1

[Signature]
Panel Member 2

V. Arun 12/12/23
Domain Coordinator

[Signature]
HOD (MECH)

RECORD OF CLASS WORK

Date	Period	Cumulative Hours	Topics Covered	Instructional Methods	Initial
			<u>UNIT-I VEHICLE STRUCTURE AND ENGINES</u>		
24.07.23	3	1	Introduction to Automobile vehicle & Types	Active	
25.07.23	1	2	Vehicle Construction & different layout.	Active	L.H.H.
26.07.23	5	3	Chassis - Requirement, Classification and different layout.	Active	L.H.H.
28.07.23	7	4	REFLECTIVE JOURNAL ACTIVITY - LIBRARY	Active	L.H.H.
31.07.23	3	5	Frame - Sections, Material, Defects & Loads on frame.	Active	L.H.H.
01.08.23	1	6	Frame - Construction, Members & types	Active	L.H.H.
02.08.23	5	7	Types of bodies, body work, Construction & Car body style.	Active	L.H.H.
04.08.23	7	8	Vehicle aerodynamics - Effects of side winds, Cross wind exposure & Vehicle veering & Dynamics of vehicle.	Passive	L.H.H.
05.08.23	3	9	IC engines - Components - Functions & Materials	Passive	L.H.H.
07.08.23	3	10	VVT - Changing of Cam, Phasing of Cam VTEC / Review of Unit - 2.	Active	L.H.H.
16.08.23	7	14	REFLECTIVE JOURNAL ACTIVITY - PRESENTATION & EVALUATION AS PER RUBRICS (ACTIVITY 2)	Active	L.H.H.

16/8/23